

CODE	COURSE NAME	CATEGORY	L	T	P	CREDIT
222ECS016	KNOWLEDGE BASED SYSTEM DESIGN	PROGRAMME ELECTIVE 3	3	0	0	3

Preamble: This course helps to familiarize with the architecture and design principles of knowledge-based systems. It covers Knowledge Acquisition, Knowledge Representation and the development of Knowledge-Based Systems. On successful completion of the course, the students will be able to distinguish between human and artificial experts, their problem-solving strategies, and their knowledge representation methods.

Course Outcomes: After the completion of the course the student will be able to

CO 1	Understand the fundamentals of knowledge engineering (Cognitive Knowledge Level: Understand)
CO 2	Apply the concepts of problem solving for building Expert System (Cognitive Knowledge Level: Apply)
CO 3	Apply the concepts of Knowledge based system for developing applications. (Cognitive Knowledge Level: Apply)
CO 4	Apply knowledge representation methods for developing Knowledge based system (Cognitive Knowledge Level: Apply)
CO 5	Make use of Semantics of Expert Systems for building applications. (Cognitive Knowledge Level: Apply)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

PO4: An ability to apply stream knowledge to design or develop solutions for real world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tool to model, analyse and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1							
CO 2							
CO 3							
CO 4							
CO 5							

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	40
Analyse	20
Evaluate	
Create	

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

Evaluation shall only be based on application, analysis or design based questions (for both internal and end semester examinations).

Continuous Internal Evaluation: 40 marks

- Preparing a review article based on peer reviewed original publications (minimum 10 publications shall be referred) : 15 marks
- Course based task / Seminar/ Data collection and interpretation : 15 marks

iii. Test paper (1 number) : 10 marks

Test paper shall include minimum 80% of the syllabus.

Course based task/test paper questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation and understanding of the students.

End Semester Examination Pattern:

The end semester examination will be conducted by the respective College.

There will be two parts; Part A and Part B.

Part A will contain 5 numerical/short answer questions with 1 question from each module, having 5 marks for each question. Students should answer all questions. Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem solving and quantitative evaluation), with minimum one question from each module of which student should answer any five. Each question can carry 7 marks

Total duration of the examination will be 150 minutes.

Note: The marks obtained for the ESE for an elective course shall not exceed 20% over the average ESE mark % for the core courses. ESE marks awarded to a student for each elective course shall be normalized accordingly.

For example if the average end semester mark % for a core course is 40, then the maximum eligible mark % for an elective course is $40+20 = 60\%$.

Course Level Assessment Questions

Course Outcome 1 (CO1):

1. Describe the role of inference engine in Rule-based expert system.
2. Illustrate the difference between an expert system and a knowledge based system?

Course Outcome 2 (CO2)

1. Explain conflict resolution technique in rule based systems.
2. Compare Hayes-Roth's and Clancey's classification of expert system

Course Outcome 3(CO3):

:COMPUTER SCIENCE AND ENGINEERING-CS2

1. Illustrate the advantages and limitations of a Knowledge based system.
- 2.. Describe the components of knowledge base system and illustrate the need of each component with an wexample.

Course Outcome 4 (CO4):

1. Discuss knowledge acquisition strategies in MUD system
2. Differentiate between procedural knowledge and factual knowledge

Course Outcome 5 (CO5):

1. What makes case-based reasoning different from rule based reasoning? Which one will be better for knowledge based system design?
2. What are the sources of uncertain reasoning?

Model Question Paper		:COMPUTER SCIENCE AND ENGINEERING-CS2	
QP CODE:			
Reg No: _____			
Name: _____		PAGES :	
4			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
SECOND SEMESTER M.TECH DEGREE EXAMINATION, MONTH & YEAR			
Course Code:			
Course Name:			
Max. Marks : 60		Duration: 2.5 Hours	
Answer All Questions. Each Question Carries 5 Marks			
1	How does an expert system help in solving complex problems?		
2	Why don't rule based systems backtrack much?		
3	Why don't all expert systems use Bayes' Rule to compute the degrees of certainty associated with conclusions? Is there any other method to solve such problems?		
4	Why is Knowledge acquisition a bottle neck? What solutions have been proposed?		
5	What is a case library and how does a case differ from a book?		(5x5=25)
Part B (Answer any five questions. Each question carries 7 marks)			
6	(a)	What are the characteristics and components of an expert system?	(3)
	(b)	Differentiate 'search space' and 'solution space'. What will be the search space and solution space for the game of chess?	(4)

7	(a)	How can we select a knowledge engineering tool? How hard such tools are to learn and use?	(4)
	(b)	List out the classification of expert system tasks.	(3)
8.		Describe the Knowledge-Based System architecture and its working.	(7)
9.	(a)	Identify some knowledge acquisition strategies for heuristics classification.	(3)
	(b)	What all difficulties may arise in the implementation phase of building a Knowledge-Based System?	(4)
10.		When the search space is potentially large, which type of testing is beneficial? Explain how it works?	(7)
11.	(a)	Describe the working of any industrial or commercial expert systems.	(4)
	(b)	What do you mean by Knowledge acquisition? What are the stages of knowledge acquisition?	(3)
12.		What are the common pitfalls during Knowledge Based Systems implementation and how we can avoid them?	(7)

Syllabus

	Syllabus	
Module	Content	Hours
I	Introduction to Knowledge Engineering: The Human Expert and an Artificial Expert –Knowledge based systems architecture-Types of Knowledge- Characteristics of Knowledge-Components of Knowledge-Basic structure of knowledge based system- Knowledge Base -Inference Engine	9
II	Problem Solving Process- Rule Based Systems – Heuristic Classifications – Constructive Problem Solving – Toolsfor Building Expert Systems - Expert System Architectures	6
III	Knowledge Based Systems: Objectives of KBS - Components of KBS - Categories of KBS – Difficulties- KBS Architecture: Basic Structure-Applications	7

IV	Developing Knowledge-Based Systems: Difficulties - Development Model - Knowledge Acquisition - Developing Relationships with Experts - Sharing Knowledge - Dealing with Multiple Experts - Knowledge Representation-Factual Knowledge - Procedural Knowledge, Knowledge Management	10
V	Case Based Reasoning – Semantics of Expert Systems – Modelling of Uncertain Reasoning. Machine Learning – Rule Generation and Refinement – Learning Evaluation – Testing and Tuning.	7

Course Plan

No	Topic	No. of Lectures (39)
1	Module 1 (Knowledge Engineering)	
1.1	Introduction to Knowledge Engineering	1
1.2	The Human Expert and an Artificial Expert	1
1.3	Knowledge based systems architecture	1
1.4	Types of Knowledge	1
1.5	Characteristics of Knowledge	1
1.6	Components of Knowledge	1
1.7	Basic Structure of Knowledge-Based Systems	1
1.8	Knowledge Base	1
1.9	Inference Engine	1
2	Module 2 (Building Expert Systems)	
2.1	Problem Solving Process	1
2.2	Rule Based Systems	1
2.3	Heuristic Classifications	1
2.4	Constructive Problem Solving	1
2.5	Tools for Building Expert Systems	1
2.6	Expert System Architectures	1
3	Module 3 (Knowledge Based Systems)	
3.1	Knowledge Based Systems	1
3.2	Objectives	1
3.3	Components of Knowledge based system	1
3.4	Categories of Knowledge based system	1
3.5	Difficulties	1
3.6	Basic Structure of KBS	1
3.7	Applications of KBS	1
4	Module 4 (Development of Knowledge-Based Systems)	
4.1	Developing Knowledge-Based Systems: Difficulties	1

4.2	Development Model	1
4.3	Knowledge Acquisition	1
4.4	Developing Relationships with Experts	1
4.5	Sharing Knowledge	1
4.6	Dealing with Multiple Experts	1
4.7	Knowledge Representation	1
4.8	Factual Knowledge	1
4.9	Representing Procedural Knowledge	1
4.10	Knowledge management	1
5	Module 5 (Case Based Reasoning)	
5.1	Case Based Reasoning	1
5.2	Semantics of Expert Systems	1
5.3	Modelling of Uncertain Reasoning.	1
5.4	Machine Learning	1
5.5.	Rule Generation	1
5.6	Refinement Learning Evaluation	1
5.7	Testing and Tuning.	1

Reference Books

1. Peter Jackson. Introduction to Expert Systems. 3 rd Edition, Pearson Education 2007.
2. Rajendra Arvind Akerkar . Knowledge-Based Systems. Jones and Bartlet Publishers 2010
3. Robert I. Levine, Diane E. Drang, Barry Edelson. AI and Expert Systems: a comprehensive guide, C language, 2nd edition, McGraw-Hill
4. Jean-Louis Ermine: Expert Systems: Theory and Practice. 4th printing, Prentice-Hall of India, 2001
5. Stuart Russell, Peter Norvig: Artificial Intelligence: A Modern Approach. 2nd Edition
6. N.P.Padhy: Artificial Intelligence and Intelligent Systems. 4th impression , Oxford University Press, 2007.